

SoftwareSpecification.md – DcSim v0.3

Purpose

This document outlines the functional and non-functional requirements for the DcSim railway DC traction power simulation software. It references the **USER_GUIDE.md** for all detailed configuration structures and examples to avoid duplication.

1. Scope

DcSim simulates DC railway traction systems, including:

- Moving trains following defined timetables and load profiles.
- Electrical grid models with nodes, lines, substations, and connection points.
- Power flow calculations at each simulation tick.

2. References

- **USER_GUIDE.md** – Contains detailed configuration syntax, examples, and parameter explanations.
- **README.md** – Provides developer instructions.

3. High-Level Functional Requirements

1. Load configuration from `application.conf`.
2. Initialize track, grid, traffic, and power profile subsystems.
3. Run simulations according to `simulationControl` parameters.
4. Output results for analysis and optional plotting.

4. Simulation Workflow

1. Parse `application.conf` (see **USER_GUIDE** for structure).
2. Create models:
 - Track stations and positions.
 - Electrical grid with nodes, lines, substations, connection points.
 - Traffic (train timetable and templates).
 - Power profiles per train.
3. Time iteration:
 - At each tick (from `tickDuration`), update train positions and loads.

- Solve the electrical system.
- Store results.

5. Non-Functional Requirements

- Java 17+, Scala 2.13, sbt.
- Compatible with `.xlsx` load profile files.
- Modular architecture for extension.

6. Constraints

- Large datasets must be handled efficiently.
- Accurate time-step simulation.

7. Notes

- For configuration examples and field-level descriptions, **always refer to USER_GUIDE.md**.
- Any change in configuration format must first be updated in USER_GUIDE.md.